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Revision History

Version	Changes
2.0	Added Version Number to document. Added sections on Default Signal Values for unconnected ports.
2.1	Added PLL primitives
2.2	Corrected SB_CARRY connections to LUT inputs
2.3	Added iCE40 RAM, PLL primitives.
2.4	Added PLL_DS, SB_MIPI_RX_2LANE, SB_TMDS_deserializer primitives.
2.5	Added SB_MAC16 Primitive details.
2.6	Added iCE40LM Hard Macro details. Removed PLL_DS, SB_MIPI, SB_TMDS, SB_MAC16 primitive details.
2.7	Added iCE5LP (iCE40 Ultra) primitive details.
2.8	Removed iCE65 RAM, PLL details.
2.9	Added iCE40UL (iCE40 Ultra Lite) primitive details
3.0	Added iCE40UP (iCE40 Ultra Plus) primitives.

Table of Contents

Register Primitives 7
SB_DFF 7
SB_DFFE 9
SB_DFFSR 11
SB_DFFR 13
SB_DFFSS 15
SB_DFFS 17
SB_DFFESR 19
SB_DFFER 21
SB_DFFESS 23
SB_DFFES 25
SB_DFFN 27
SB_DFFNE 29
SB_DFFNSR 31
SB_DFFNR 33
SB_DFFNSS 35
SB_DFFNS 37
SB_DFFNESR 39
SB_DFFNER 41
SB_DFFNESS 43
SB_DFFNES 45
Combinational Logic Primitives 47
SB_LUT4 47
SB_CARRY 49
Block RAM Primitives 51
iCE40 Block RAM 51
SB_RAM256x16 52
SB_RAM256x16NR 54
SB_RAM256x16NW 55
SB_RAM256x16NRNW 57
SB_RAM512x8 60
SB_RAM512x8NR 62
SB_RAM512x8NW 63
SB_RAM512x8NRNW 65
SB_RAM1024x4 68
SB_RAM1024x4NR 70
SB_RAM1024x4NW 71
SB_RAM1024x4NRNW 73
SB_RAM2048x2 76
SB_RAM2048x2NR 78
SB_RAM2048x2NW 79
SB_RAM2048x2NRNW 81

SB_RAM40_4K 83
IO Primitives 88
SB_IO 88
Global Buffer Primitives 92
SB_GB_IO 92
SB_GB Primitive 93
PLL Primitives 94
iCE40 PLL Primitives 94
SB_PLL40_CORE 94
SB_PLL40_PAD 98
SB_PLL40_2_PAD 102
SB_PLL40_2F_CORE 105
SB_PLL40_2F_PAD 109
Hard Macro Primitives 113
iCE40LM Hard Macros 113
SB_HSOSC (For HSSG) 113
SB_LSOSC (For LPSG) 114
SB_I2C 114
SB_SPI 117
iCE5LP (iCE40 Ultra) Hard Macros 121
SB_HFOSC 121
SB_LFOSC 122
SB_LED_DRV_CUR 123
SB_RGB_DRV 124
SB_IR_DRV 125
SB_RGB_IP 127
SB_IO_OD 128
SB_I2C 130
SB_SPI 131
SB_MAC16 132
iCE40UL (iCE40 Ultra Lite) Hard Macros 145
SB_HFOSC 145
SB_LFOSC 146
SB_RGBA_DRV 147
SB_IR400_DRV 149
SB_BARCODE_DRV 150
SB_IR500_DRV 151
SB_LEDDA_IP 153
SB_IR_IP 154
SB_IO_OD 157
SB _I2C_FIFO 159
iCE40UP (iCE40 Ultra Plus) Hard Macros 162
SB_HFOSC 162

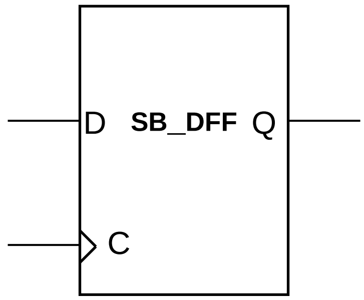
SB_LFOSC 163
SB_RGBA_DRV 164
SB_LEDDA_IP 166
SB_IO_OD 167
SB_I2C 170
SB_SPI 171
SB_MAC16 172
SB_SPRAM256KA 173
SB_IO_I3C 176
Device Configuration Primitives 180
SB_WARMBOOT 180

Register Primitives

SB_DFF

D Flip-Flop

Data: D is loaded into the flip-flop during a rising clock edge transition.



Inputs			Output
	D	**C**	**Q**
	0	↑	0
	1	↑	1
Power on State	X	X	0

Key
↑ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL use

This register is inferred during synthesis and can also be explicitly instantiated.

Verilog Instantiation

// SB_DFF - D Flip-Flop.

```
SB_DFF SB_DFF_inst (
       .Q(Q),             // Registered Output
       .C(C),             // Clock
       .D(D),             // Data
       );
```

// End of SB_DFF instantiation

VHDL Instantiation

-- SB_DFF - D Flip-Flop.

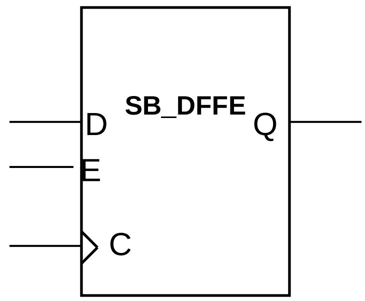
```
SB_DFF_inst: SB_DFF
      port map (
      Q => Q,            -- Registered Output
      C => C,            -- Clock
      D => D,            -- Data
      );
```

-- End of SB_DFF instantiation

SB_DFFE

D Flip-Flop with Clock Enable

Data D is loaded into the flip-flop when Clock Enable E is high, during a rising clock edge transition.



Inputs			Output
E	**D**	**C**	**Q**
0	X	X	Previous **Q**
1	0	↑	0
1	1	↑	1
Power on State	X	X	0

Key
↑ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input E: Logic '1'

Note that explicitly connecting a logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the logic '1'. It is recommended that the user leave the port E unconnected, or use the corresponding flip-flop without Enable functionality i.e. the DFF primitive.

Verilog Instantiation

// SB_DFFE - D Flip-Flop with Clock Enable.

```
SB_DFFE      SB_DFFE_inst (
             .Q(Q),              // Registered Output
             .C(C),              // Clock
             .D(D),              // Data
             .E(E),              // Clock Enable
             );
```

// End of SB_DFFE instantiation

VHDL Instantiation

-- SB_DFFE - D Flip-Flop with Clock Enable.

```
SB_DFFE_inst: SB_DFFE
            port map (
            Q => Q,             -- Registered Output
            C => C,             -- Clock
            D => D,             -- Data
            E => E,             -- Clock Enable
            );
```

-- End of SB_DFFE instantiation